```sas
/*
Programmed by: Thomas M. Grover
Programmed on: 2025-04-15
Programmed to: Complete FP02
Modified by: N/A
Modified on: N/A
Modified to: N/A
*/

x "cd L:\st445\Data\BookData\BeverageCompanyCaseStudy";
filename RawData ".";

x "cd L:\st445\Results\FinalProjectPhase1";
libname InputDS ".";

x "cd S:\Documents\FP02";
libname Exam ".";

*Set options;
options nodate;
ods noproctitle;
ods listing close;
ods graphics on / width = 6in;

*Set output;
ods pdf file = "GroverFinalReport.pdf" dpi = 300
                                       style = sapphire;
ods pdf exclude all;


*2.1;
ods pdf select all;
ods escapechar = "^";
title j=c "Activity 2.1^ Summary of Units Sold^ Single Unit Packages";
footnote h=8pt "Minimum and maximum Sales are within any county for any week";
proc means data = InputDS.fp01dugginsalldata(
    keep = StateFips ProductName Size UnitSize UnitsSold ProductCategory
    where = (ProductCategory = "Soda: Cola" and UnitSize = 1 and StateFips in (13, 37, 45)))
    sum min max nonobs;
    class StateFips ProductName Size UnitSize;
    var   UnitsSold;
run;
title "";
footnote "";

*2.3;
ods escapechar = "^";
title j=c "Activity 2.3^ Cross Tabulation of Single Unit Product Sales in Various States";
ods pdf select Freq.Table1of1.CrossTabFreqs Freq.Table2of1.CrossTabFreqs;
proc freq data = InputDS.fp01dugginsalldata;
    where  ProductCategory = "Soda: Cola" and StateFips in (13, 37, 45);
    tables ProductName*StateFips*Size;
run;
title "";

*3.1;
ods pdf select all;
ods escapechar = "^";
title h=12pt "Activity 3.1";
title2 j=c "Single-Unit 12 oz Sales^ Regular, Non-Cola Sodas";
proc sgplot data = InputDS.fp01dugginsalldata(
    where = (StateFips in (13, 37, 45) and
             ProductCategory = "Soda: Non-Cola" and
             Type = "Non-Diet" and
             UnitSize = 1 and
             Size = "12 oz"));
    hbar StateName / response     = UnitsSold
                     stat         = sum
                     group        = ProductName
                     groupdisplay = cluster;
    keylegend / title    = ""
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                location = inside
                position = SE
                down     = 3;
    xaxis label   = "Total Sold";
    yaxis display = (nolabel);
run;
title "";

*3.3;
title h=12pt "Activity 3.3";
title2 j=c "Average Weekly Sales, Non-Diet Energy Drinks^ For 8 oz Cans in Georgia";
proc sgplot data = InputDS.fp01dugginsalldata(
    where = (StateName       = "Georgia" and
             Type            = "Non-Diet" and
             ProductCategory = "Energy" and
             Size            = "8 oz"));
    vbar ProductName / response     = UnitsSold
                       stat         = mean
                       group        = UnitSize
                       groupdisplay = cluster
                       dataskin     = Sheen;
    xaxis display = (nolabel);
    yaxis label   = "Weekly Average Sales";
run;
title "";

*3.6;
title h=12pt "Activity 3.6";
title2 j=c "Weekly Average Sales, Nutritional Water^ Single-Unit Packages";
proc sgplot data = InputDS.fp01dugginsact3_6results;
    hbar ProductName / response    = UnitsSold_Mean
                       barwidth    = .48
                       legendlabel = "Mean Sales";
    hbar ProductName / response    = UnitsSold_Median
                       barwidth    = .8
                       fillattrs   = (transparency = 0.4)
                       legendlabel = "Median Sales";
    xaxis label = "Georgia, North Carolina, and South Carolina";
    yaxis display = (nolabel);
    keylegend / title    = "Weekly Sales"
                location = inside
                position = NE
                down     = 2
                noborder;
run;
title "";

*4.1;
title j=c "Activity 4.1^ Weekly Sales Summary^ Cola Products, 20 oz Bottles, Individual Units";
footnote h=8pt "All States";
proc means data = InputDS.fp01dugginsalldata
           mean median q1 q3 nonobs
           maxdec = 0;
    where ProductCategory = "Soda: Cola" and
          Size            = "20 oz" and
          UnitSize        = 1;
    class Region Type Flavor;
    var   UnitsSold;
run;
title "";
footnote "";

*4.2;
title h=12pt "Activity 4.2";
title2 j=c "Weekly Sales Distributions^ Cola Products, 12 Packs of 20 oz Bottles";
footnote h=8pt "All States";
proc sgpanel data = InputDS.fp01dugginsalldata(
    where    = (ProductCategory = "Soda: Cola" and
    Size     = "20 oz" and
    UnitSize = 12));
    panelby Region Type / columns = 2
                          novarname;
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    histogram UnitsSold / scale = percent
                          binwidth = 250;
    colaxis label   = "Units Sold";
    rowaxis display = (nolabel);
    format UnitsSold percent7.;
run;
title "";
footnote "";

*4.4;
title h=12pt "Activity 4.4";
title2 j=c "Sales Inter-Quartile Ranges^ Cola: 20 oz Bottles, Individual Units";
footnote h=8pt "All States";
proc sgpanel data = InputDS.fp01dugginsact4_4results;
  panelby Region Type / novarname;
  highlow x = Date
          low = UnitsSold_Q1
          high = UnitsSold_Q3 / lineattrs = (thickness = 1 color = darkblue);
  colaxis interval     = month
          valuesformat = MONYY7.
          label        = "Date";
  rowaxis label        = "Q1-Q3";
run;
title "";
footnote "";

*Optional;
title j=c "Optional Activity^ Product Information and Categorization";
proc report data = InputDS.fp01dugginsclassification;
    columns ProductName Type ProductCategory ProductSubCategory Flavor Size Container;
run;
title "";

*5.5;
title h=12pt "Activity 5.5";
title2 j=c "North and South Carolina Sales in August^ 12 oz, Single-Unit, Cola Flavor";
proc sgpanel data = InputDS.fp01dugginsact5_5trans;
    attrib date format = mmddyy8.;
    panelby Type / columns     = 1
                   novarname;
    hbar Date    / response    = North_Carolina
                   barwidth    = .48
                   name        = "NC"
                   legendlabel = "North Carolina";
    hbar Date    / response    = South_Carolina
                   barwidth    = .8
                   fillattrs   = (transparency = 0.4)
                   name        = "SC"
                   legendlabel = "South Carolina";
    colaxis label        = "Sales"
            valuesformat = comma7.
            type         = linear;
    rowaxis display      = (nolabel);
    keylegend "NC" "SC";
run;
title "";

*6.2;
title j=c "Activity 6.2^ Quarterly Sales Summaries for 12oz Single-Unit Products^ Maryland Only";
proc report data = InputDS.fp01dugginsalldata(
    where = (StateName = "Maryland" and
             Size = "12 oz" and
             UnitSize = 1))
    style(summary) = [color = black backgroundcolor = white];
    column Type ProductName Date UnitsSold = MedianWeeklySales
                                 UnitsSold = TotalSales
                                 UnitsSold = LowestWeeklySales
                                 UnitsSold = HighestWeeklySales;
    define Type               / group;
    define ProductName        / group;
    define Date               / group
                                format=QTRR.
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                                "Quarter"
                                 order = internal;
    define MedianWeeklySales  / analysis median
                               "Median Weekly Sales";
    define TotalSales         / analysis sum
                                 format = comma10.
                               "Total Sales";
    define LowestWeeklySales  / analysis min
                               "Lowest Weekly Sales";
    define HighestWeeklySales / analysis max
                               "Highest Weekly Sales";
    break after ProductName   / summarize suppress;
run;
title "";


data Exam.Sodas;
    infile RawData("Sodas.csv") firstobs = 6 dlm = "," truncover;
    input ProductNumber : 8.
          ProductName : $50. @ ;
    length ProductNumber 8 ProductName $50 Size $20 UnitSize 8 ProductCode $50;
    do i = 1 to 6;
        input _holder : $50. @ ;
        if missing(_holder) then leave;
        _sizeval = scan(_holder, 1, " ");
        _sizeunit = scan(_holder, 2, " ");
        Size = catx(" ", _sizeval, _sizeunit);
        _multiunits = index(_holder, "(");
        if _multiunits > 0 then do;
            _pull = substr(_holder, (_multiunits+1), length(_holder)-(_multiunits-1));
        end;
        else do;
            _pull = '1';
        end;
        do w = 1 to countw(_pull, ',');
            UnitSize = strip(scan(_pull, w, ','));
            ProductCode = catx('-', 'S', Size, UnitSize);
            output;
        end;
    end;
    attrib ProductNumber label = "Product Number"
           ProductName   label = "Product Name"
           Size          label = "Beverage Volume"
           UnitSize      label = "Beverage Quantity"
                         format = best12.
           ProductCode   label = "Product Code";
    drop _holder _sizeval _sizeunit _multiunits _pull i w;
run;

*7.1;
proc sort data = Exam.Sodas nodupkey;
    by ProductNumber ProductName Size;
run;

*7.4;
title j=c "Activity 7.4^{newline} Quarterly Sales Summaries for 12oz Single-Unit Products^{newline} Maryland Only";
proc report data = InputDS.fp01dugginsalldata(
    where = (StateName = "Maryland" and
             Size = "12 oz" and
             UnitSize = 1))
    style(header) = [color = cx89CFF0 backgroundcolor = cx778899]
    style(summary) = [color = white backgroundcolor = black];
    column Type ProductName Date UnitsSold = MedianWeeklySales
                                 UnitsSold = TotalSales
                                 UnitsSold = LowestWeeklySales
                                 UnitsSold = HighestWeeklySales;
    define Type               / group;
    define ProductName        / group;
    define Date               / group
                                format=QTRR.
                                  "Quarter"
                                  order = internal;
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    define MedianWeeklySales  / analysis median
                                "Median Weekly Sales";
    define TotalSales         / analysis sum
                                format = comma10.
                                "Total Sales";
    define LowestWeeklySales  / analysis min
                                "Lowest Weekly Sales";
    define HighestWeeklySales / analysis max
                                "Highest Weekly Sales";
    compute before ProductName;
        rowCounter = 0;
    endcomp;
    compute Date;
        if _break_ = "" then do;
            rowCounter + 1;
            select (mod(rowCounter, 4));
                when (1) call define(_row_, "style", "style = [backgroundcolor = white]");
                when (2) call define(_row_, "style", "style = [backgroundcolor = cxE0E0E0]");
                when (3) call define(_row_, "style", "style = [backgroundcolor = cxC0C0C0]");
                when (0) call define(_row_, "style", "style = [backgroundcolor = cxA0A0A0]");
            end;
        end;
    endcomp;
    break after ProductName / summarize suppress;
run;
title "";

*7.5;
title j=c "Activity 7.5^ Quarterly Per-Capita Sales Summaries^ 12oz Single-Unit Lemonade^ Maryland Only";
footnote h=8pt "Flagged Rows: Sales Less Than 7.5 per 1000 for Diet; Less Than 30 per 1000 for Non-Diet";
proc report data = InputDS.fp01dugginsalldata(
    where = (StateName = "Maryland" and
             Size = "12 oz" and
             UnitSize = 1 and
             index(ProductName,"Lemonade") > 0))
    style(header) = [color = cx89CFF0 backgroundcolor = cx778899]
    style(summary) = [color = white backgroundcolor = cx363737];
    column CountyName Type Date UnitsSold = TotalSales SalesPerThousand PopEstimate2016 PopEstimate2017;
    define CountyName / group;
    define Type       / group;
    define Date       / group
                        format=QTRR.
                        "Quarter"
                        order = internal;
    define TotalSales       / analysis sum
                              format = comma10.
                              "Total Sales";
    define SalesPerThousand / analysis sum
                              format = 7.1
                              "Sales per 1,000";
    define PopEstimate2016  / analysis mean
                              format = comma12.1
                              noprint;
    define PopEstimate2017  / analysis mean
                              format = comma12.1
                              noprint;
    break after CountyName  / summarize suppress;
    compute CountyName;
        CountyName = tranwrd(CountyName, " County", "");
    endcomp;
    compute after CountyName / style = [color = white backgroundcolor = black];
        MeanPop = (PopEstimate2016.mean + PopEstimate2017.mean) / 2;
        line "Average 2016 & 2017 Population:" MeanPop comma10.1;
    endcomp;
    compute SalesPerThousand;
        if SalesPerThousand.sum < 7.5 then do;
            call define(_col_, "style", "style = [color = red]");
            call define(_row_, "style", "style = [backgroundcolor = cxC0C0C0]");
        end;
        else if (Type = "Non-Diet" and SalesPerThousand.sum < 30) then do;
            call define(_col_, "style", "style = [color = red]");
            call define(_row_, "style", "style = [backgroundcolor = cxC0C0C0]");
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        end;
    endcomp;
run;
title "";
footnote "";


*Standard end of code;
ods pdf close;
ods listing;
```